# Methodology: gauge-free reservoir storage monitoring and forecasting

DATA

**Sen1Floods11**

Hand-labelled water chips
Sentinel-1 VV/VH +
Sentinel-2 (13 bands)
Train / val / test +
held-out Bolivia

**Sentinel-1 & Sentinel-2**

2017–2020, Earth Engine
S1 GRD (track 92, daily
mosaics, ~12-day repeat)
S2 L1C + s2cloudless
cloud probability

**Reservoir outlines**

JRC Global Surface
Water max extent
Connected component
+ 300 m buffer
4 Chennai reservoirs

**Hydrological data**

Daily gauge storage &
rainfall, 2004–2020
(Chennai Water Mgmt.)
CHIRPS rainfall,
ERA5-Land evaporation

MODELS & PREP.

**Water segmentation (U-Net, ResNet-18)**

S2-only | S1-only | fused S1+S2
Fused: with / without robustness training
(modality dropout + synthetic clouds)
Loss BCE + Dice; MC-dropout uncertainty
Baselines: NDWI, MNDWI, S1 Otsu, −16 dB
Metrics: IoU, F1, ECE, IoU vs cloud (0–100 %)
on test and held-out Bolivia splits
Free GPU (Kaggle T4), seeds 0 and 1

*imagery + outlines*

**Gauge data preparation**

Quality control: 30 isolated spike values
(1–3-day blocks) removed, interpolated
Storage normalised by reservoir capacity
Chronological splits: train 2004–14 |
calibration 2015–16 | test 2017–20
Cholavaram gauge flagged (flat record,
disagrees with satellite): not used for
fitting, reported separately

*trained models*

WATER MAPPING

**Reservoir water area time series (images streamed from Earth Engine, no export)**

Every scene on a 10 m UTM grid; six methods on identical pixels:
S2 U-Net, S1 U-Net, fused U-Net, MNDWI, NDWI, S1 Otsu
Cloud fraction from s2cloudless (> 50 %); optical methods kept only if < 10 % cloud over the reservoir
Sensor switching: S2 U-Net on clear dates, S1 U-Net on all other dates (cloud-independent, ~12-day revisit)
Output: water area (km²) per reservoir, date and method, with MC-dropout spread

*gauge storage & rainfall*

*water area*

STORAGE & FORECASTS

**Area → storage (rating curves)**

Power law S = a + b·A^c
Gauged: own gauge record,
time-blocked cross-validation
Ungauged (gauge-free): curve fitted
on OTHER reservoirs, S/capacity
vs A/max area (leave-one-out)
Dry-bed false-water check
(area when storage < 5 % capacity)

*storage*

**Storage forecasting (7, 15, 30 days)**

Inputs: storage lags (0–60 d), changes, rainfall
sums (1–90 d), seasonality, days since last obs.
Models: persistence, climatology, AR-ridge,
random forest, HistGB, MLP
Satellite-like training inputs (revisit, gaps, noise)
90 % intervals: split, Mondrian (season × reservoir),
adaptive conformal; quantile regression
Gauge-free: forecaster trained on other reservoirs
+ CHIRPS rainfall + satellite storage

EVALUATION

**Evaluation against gauge storage**

Storage and forecasts: NSE, KGE, RMSE, MAE, PBIAS; Diebold–Mariano test vs persistence; by season (dry, SW, NE monsoon)
Intervals: coverage (PICP), width (MPIW), Winkler score; transfer to unseen reservoirs (leave-one-reservoir-out)
Error budget: daily gauge vs satellite dates with exact storage vs satellite-derived storage